Papers 9–13 Creation, Review, and Integrity Process Record

English edition | ARS Stage 6

Date:	16 August 2026 (Asia/Shanghai)
Scope:	Papers 9, 10, 11, 12, and the Paper 13 Technical Note
Technical verdict:	**PASS —C0/M0/m0**
Public release:	`PUBLIC_RELEASE_AUTHORIZED=false`
Stage 6:	`in_progress`, awaiting terminal acknowledgement

Reconstructed from exact-byte artifacts, append-only reviews, and the batch audit.
This record is not an authorship declaration, submission authorization, or public-release permit.

Contents

1 Executive summary

Papers 9–13 have completed research design, source and ownership audits, mathematical proofs, deterministic controls, Route evaluation, composition, citation preflight, manuscript review, and technical release audit. The five exact six-file packages total 89 pages. Batch controls comprise 399/399 tests, 53 CSV files, 7,709 body rows, and 86 explicit negatives. Of 40 Route-A records, 25 are exploratory and 15 rejected; every A2–A4 coordinate fails, and there are no Route-B files.

Papers 9–12 retain article disposition. Paper 13 is internally accepted as a Technical Note. Its independent substantive-weight result remains `NOTE_OR_MERGE`, `STANDALONE_PASS=false`, and `C0/M1/m0`. This record does not erase that Major; it records it as the reason for the Technical Note disposition.

The primary upstream receipt is:

- `papers/9-packet-separation/notes/papers9_13_batch_audit.md`
- SHA-256: `6aa915a9e85153957b269448ba23b56716c4f64d18e6b3c85f904d73b0001aea`
- Current provenance graph: 65 nodes, 255 edges, no self-edge, and no cycle.

2 Evidence boundary and attribution

The early raw dialogue is unavailable after long-session compaction. The initial prompt, turn numbers, exact human-intervention count, and complete concession metrics cannot be reconstructed honestly. The only currently visible user text safe to quote verbatim is:

“继续”(“continue”)

That message followed delivery of the Stage-5 five-paper batch audit and is treated as explicit consent to enter Stage 6. Earlier scope is described only as artifact-backed inference, never as a fabricated quotation.

The compacted session summary—not a surviving verbatim turn—records the five-paper scope, broad automation, and the no-Git/no-public-sync constraint before audit closure; frozen artifacts corroborate the batch scope and release hold. “继续”remains the only currently visible user text quoted verbatim. Proof writing, source verification, defect discovery, remediation, and technical closure are attributed to the AI/review workflow unless a fuller raw transcript later establishes otherwise.

3 Final position of the five papers

Paper	Final position	Pages	PDF SHA-256 (abbreviated)
P9	internally accepted article package	21	`c55e4f45...85e02`
P10	internally accepted article package	19	`30c22eb8...2ce4`
P11	internally accepted article package	16	`15d20756...840d`
P12	internally accepted; `STANDALONE_PASS` retained	18	`9d6747e9...c15`
P13	internally accepted Technical Note	15	`4082ca13...43c2`

Each `paper/` directory contains exactly six ordinary files: README, TeX, BibTeX, two native TikZ sources, and one PDF. No package contains a symlink, build auxiliary, second PDF, or research-source PDF.

4 Evidence-aligned stage chronology

Stage	Input	Output and decision	Iteration, correction, and final gate
1. Batch intake and freeze	P9–P13 protocols, candidate locks, predecessor boundaries	Five owner-typed projects; delivery consolidated at the batch boundary	The original prompt is unavailable. Historical `pipeline_state.md` files remain gate-time snapshots.
2. Phase-1 design relock	Methodology, devil/domain, and source-feasibility reviews	All five Phase-1 packages ultimately closed	P9 two states; P10 two; P11 two; P12 initial/v1/v2; P13 initial/v1; all final `C0/M0/m0`.
3. Phase-2 source and precedent work	Retained PDFs, owner dictionaries, bounded searches	Source, terminology, ownership, and novelty ceilings	Searches retained `SUPPORTED_WITHIN_SEARCH`; no bounded search became an absolute priority claim.
4. Phase-3 proofs and controls	Source-locked claims	Stable proofs, counterexamples, and deterministic controls	P12 v2/v3/v4; P13 initial and v2 corona packages; mathematics passed and standalone was adjudicated separately.
5. P13 control remediation	First implementation and mutation probes	Replacement manifest `26a41e...094c2`	Token/self-comparison oracles caused `C0/M1/m0`; after repair: 176/176 tests, 12 CSVs, 2,665 rows, 67/67 negatives, PASS.
6. Publication disposition	Proof, source, control, and standalone reviews	P9–P12 retained article status; P13 became a Technical Note	P12 v4 closed its routine-reduction Major; P13 retained its Major under `PASS_TO_TECHNICAL_NOTE`.
7. Route evaluation	Stable proofs and control receipts	40 Route-A YAML records	25 exploratory, 15 rejected; all 120 A2–A4 coordinates FAIL; Route-B count zero.
8. Composition and citation preflight	Proof/Route tuples and source ceilings	Five composition blueprints; English bodies and independently written Simplified-Chinese abstracts	Structure, claim order, traces, and declarations were frozen first; blueprints added no mathematical evidence.
9. Manuscript review and freezes	TeX, Bib, figures, and PDFs	Five exact six-file packages	P9 initial C1/M8 then acceptance; P12 Freeze 1→2→Correction→count relock; P13 Freeze 1 C0/M2/m1→Freeze 2→status relock.
10. Bounded P12 corrections	Review Freeze 2 tuple	Corrected Bib/PDF and append-only citation→peer→release chain	Stacks title became “Colimits of spaces”; Han receipt became 353 body + 32 keywords = 385; both historical m1 findings closed.
11. Citation, peer, and release closure	Final manuscript tuples	Per-paper citation, peer, and release reports	P11–P13 append-only status relocks; all current citation graphs and technical gates PASS.

Stage	Input	Output and decision	Iteration, correction, and final gate
12. Consolidated batch audit	Five current scholarly tuples and status files	One exact-byte receipt for packages, PDFs, sources, Route, controls, and the graph	Closed the P9 report edge and P11/P12/P13 status wording; final `PASS`, `C0/M0/m0`, public release still false.

5 Minimum evidenced iteration counts

These are named artifact states, not inferred dialogue turns.

Paper	Design states	Proof/disposition states	Manuscript/finalization states
P9	2	1 stable proof package	2: initial manuscript and accepted re-review
P10	2	1 stable proof/control package	at least 2; one final spacing-only relock
P11	2	1 stable proof/control package	at least 2 citation states plus a status-relock chain
P12	3	3: v2/v3/v4	4: Freeze 1, Freeze 2, Correction, count relock
P13	2 Phase-1 and 3 control-design states	2 standalone and 2 implementation states	3: Freeze 1, Freeze 2, status relock

6 Integrity and reproducibility totals

Item	Total
Unit tests	399/399
CSVs / body rows / explicit negatives	53 / 7,709 / 86
Route-A / Route-B	40 / 0
Exploratory / rejected	25 / 15
Local research PDFs / reused manifestations	32 / 3
PDF/preflight pairs	35
Total manuscript PDF pages	89
Fonts	39, all embedded/subset/Unicode
Current artifact graph	65 nodes, 255 edges, 0 self-edge, 0 cycle

All manuscript PDFs are A4 PDF 1.5, unencrypted, attachment-free, and raster-image-free; Ghostscript parsing passes. Research PDFs remain only under `notes/sources/` and do not enter any `paper/` package.

7 Corrections that changed direction

1. **Standard ingredients cannot be repackaged as standalone work.** P12 v4's same-carrier standardization and H^1 diagonal invariant closed its earlier routine-reduction Major. After predecessor and standard-lemma subtraction, P13's corona result remained an instance of a generic isometric diagonal lemma and was routed honestly to a Technical Note.

2. **Controls must be independently fail-closed.** Correct CSV bytes did not rescue P13's first implementation when detectors merely looked up tokens or compared one formula with itself. Repair and independent reruns preceded downstream use.

3. **Owner and type checks outrank narrative convenience.** Packet transitivity, same-carrier ownership, morphism variance, and Route owners were versioned and relocked rather than concealed in prose.

4. **Metadata is integrity evidence.** P12's Stacks title and Han counting convention remain visible as historical m1 findings, then close in append-only receipts.

5. **Provenance edges must stay acyclic.** P9's `paper.log` is only a historical transient receipt. Downstream reports bind upstream bytes without self-hash or reverse-hash construction.

8 Collaboration depth trajectory (advisory)

This assessment uses Wang–Zhang rubric v1.0. It does not evaluate manuscript quality and never blocks progression. The scores are provisional with low-to-medium confidence because early raw turns are unavailable.

Dimension	Score	Result	Evidence ceiling
Delegation Intensity	8/10	high	whole categories were delegated rather than micro-edited
Cognitive Vigilance	4/10	mid-low	workflow gates were strong, but visible user challenges are absent
Cognitive Reallocation	3/10	low	no visible original synthesis, counterargument, or theorem-level judgment

The classification is **Zone 2 —Mid, automation-dominant**. Zone 3 requires all three dimensions to be at least seven; only high delegation is evidenced. Zone 1 is excluded because AI use was extensive. A future auditable Zone-3 pattern could add one human challenge to a central theorem or citation at each major gate, a written human rationale for article type and release decisions, and one original synthesis after every major correction.

9 AI self-reflection and sycophancy risk

The AI used high parallelism, exact locks, and append-only reviews. Independent reviewers repeatedly prevented premature PASS decisions. The cost was excessive status-pointer and hash-edge relocking near closure. This reflection is written by the same AI system it evaluates and is not independent evidence.

Metric	Result
DA concession rate / consecutive concessions	`UNMEASURED`; compacted logs do not preserve a reliable denominator
Skipped checkpoints	`UNMEASURED`; broad automation compressed nonmandatory prompts to the batch boundary
Visible user overrides / health alerts	0 / 0; the full health-alert history is `UNMEASURED`, and zero visible alerts do not prove no risk
Intent-mode transitions	`UNMEASURED`; the visible Stage-5-to-6 move is a stage transition, not evidence of an exploratory–goal-oriented mode change
Cross-model disagreements	cross-model was not enabled
Sycophancy risk	**UNMEASURED / requires human reading**; it cannot be labelled LOW

9.1 What the AI got wrong

- The initial P12 bibliography called Stacks Tag 0B1W "Topological colimits," and an earlier citation PASS missed the error.
- P12's Chinese count used `Script_Extensions=Han`, included 17 punctuation characters, and reported 370 instead of 353 body characters.
- P13 Freeze 1 had six-key trace, parent-README status, and Han-count defects.
- P13's first control implementation admitted tautological/token-detector oracles and required bounded remediation.
- A P12 controls audit suffered a duplicate-run orchestration incident. It caused no result drift but exposed runner-coordination weakness.
- P13 status pointers caused avoidable relock churn near closure.
- The first batch graph paragraph called the new report a sink under a `referrer->prerequisite` convention; machine review corrected it to a source.

10 Retrospective seven-mode failure audit

No uniform batch-wide contemporaneous ARS v3.20 Stage-2.5/4.5 seven-mode log survives. The final citation audits for P11–P13 contain per-paper seven-mode tables, whereas P9–P10 do not. The table below is only a Stage-6 retrospective synthesis and does not replace the missing formal records.

Mode	Retrospective assessment	History and resolution
1. Implementation bug passing self-review	CLEAR	P13 control oracles were challenged by mutation probes; replacement implementation and independent rerun closed the issue.
2. Hallucinated citation	CLEAR	Citation graphs close; P12's real-source metadata m1 was corrected and relocked.
3. Hallucinated experimental result	CLEAR	Every finite result is CSV/manifest-bound; P13's first manifest is excluded.
4. Shortcut reliance	CLEAR / theoretical boundary	Finite controls are diagnostic and never promoted to general theorems.
5. Bug reframed as insight	CLEAR	The oracle defect was repaired before downstream use, not narrated as a contribution.
6. Methodology fabrication	CLEAR	Test, row, negative, build, and PDF receipts match stable bytes.
7. Early frame-lock	CLEAR at final scope	Proposed standalone strengthenings were downgraded after counterexample or prior-art review; P13 became a Technical Note.

No user override is recorded.

11 Collaboration quality and reusable lessons

The separate legacy six-dimension scorecard gives these provisional values: Direction Setting 88, Intellectual Contribution 45, Quality Gatekeeping 78, Iteration Discipline 80, Delegation Efficiency 92, and Meta-Learning 47. The equal-weight result is **72/100 —Good**. Confidence is low-to-medium, and this score must remain distinct from the Wang–Zhang three-axis rubric.

Reusable lessons are:

- Correct artifact bytes do not imply correct control oracles; mutation probes are mandatory.
- Standalone weight must be reviewed after subtracting predecessor papers and standard lemmas.
- `pipeline_state.md` is a historical gate snapshot; later exact reports define current status.
- README, citation, peer, release, and batch reports need an acyclic binding order.
- Local source-PDF retention and public-payload exclusion are distinct gates; public safety cannot be claimed without real Git/archive/fresh-clone checks.
- Chinese count receipts must freeze the Unicode property, text boundary, and treatment of keywords.

12 External release gates and Stage-6 terminal condition

Human or real publication-system decisions remain for author order, affiliations, correspondence, ORCID, CRediT, funding, conflicts, acknowledgments, ethics/consent applicability, and final AI/tool disclosure; immutable identities or approved self-contained replacements for unpublished companions; venue and article type, template, citation style, licence, accessibility, and submission-day DOI/retraction/correction/policy checks; and real Git/index/LFS/archive/upload/attachment/hidden-path/fresh-clone source-PDF exclusion.

This record performs no Git, commit, push, archive, or upload.

Current Stage-6 status: `in_progress`

After delivery, the workflow still awaits an unambiguous user reply such as “完成”, “确认”, “结束”, “done”, or “confirm”. Before that acknowledgement, the workflow is not terminal and public-release authorization must not be inferred.